

Stage 1 IBR Version Report: IEEE 39 mix60 Case

Conversion documentation for data/processed/mix60/mix60.xlsx

Generated from conversion_protocol.yaml and ANDES workbook contents

Abstract

This document describes the generated IBR version of the IEEE 39 dynamic case, profile `mix60`. It explains which generator buses were changed, which dynamic records were removed, which ANDES IBR records were added, and which variables are copied from the original static generator data versus assigned from the configured parameter profiles. The report is about the case construction itself, not about the later time-domain comparison.

1 Case Identity

Item	Value
Protocol	<code>ieee39_ibr_conversion_protocol</code>
Active profile	<code>mix60</code>
Base workbook	<code>data/raw/ieee39_full.xlsx</code>
Generated IBR workbook	<code>data/processed/mix60/mix60.xlsx</code>
Build manifest	<code>outputs/stage1_build_ibr_case/mix60/manifest.json</code>
ANDES version in manifest	<code>2.0.0</code>
Power-flow policy	preserve the original static PV/Slack dispatch

The conversion changes the dynamic resource representation, not the solved network topology. Static network sheets such as `Bus`, `PQ`, `PV`, `Slack`, `Shunt`, `Line`, and `Area` are preserved. The static generator injections remain in `PV` and `Slack`; the replacement is made in the dynamic model sheets.

2 Profile Logic

The protocol declares buses 31–38 as eligible conversion sites and keeps buses 30, 31, 32, and 39 synchronous in `mix60`. Bus 39 is retained because it represents the external interconnection/equivalent.

Resource group	Generator indices	Bus indices
Kept synchronous	1, 2, 3, 10	30, 31, 32, 39
Converted to GFL	5, 7, 9	34, 36, 38
Converted to GFM	4, 6, 8	33, 35, 37

3 Conversion Matrix

Generator	Bus	Static record	Base dynamic records removed	New IBR records	Role
1	30	PV 1	none	none	synchronous retained
2	31	PV 2	none	none	synchronous retained
3	32	PV 3	none	none	synchronous retained
4	33	PV 4	GENROU_4, TGOV1_4, IEEEEX1_4, IEEEEST_4	REGF1_G4	GFM
5	34	PV 5	GENROU_5, TGOV1_5, IEEEEX1_5, IEEEEST_5	PLL1_G5, REGCP1_G5, REECA1_G5, REPCA1_G5	GFL
6	35	PV 6	GENROU_6, TGOV1_6, IEEEEX1_6, IEEEEST_6	REGF1_G6	GFM
7	36	PV 7	GENROU_7, TGOV1_7, IEEEEX1_7, IEEEEST_7	PLL1_G7, REGCP1_G7, REECA1_G7, REPCA1_G7	GFL
8	37	PV 8	GENROU_8, TGOV1_8, IEEEEX1_8, IEEEEST_8	REGF1_G8	GFM
9	38	PV 9	GENROU_9, TGOV1_9, IEEEEX1_9, IEEEEST_9	PLL1_G9, REGCP1_G9, REECA1_G9, REPCA1_G9	GFL
10	39	Slack 10	none	none	synchronous external equivalent

4 ANDES Sheet-Level Delta

Sheet	Base rows	mix60 rows	Meaning
GENROU	10	4	Synchronous machine dynamics retained only for generators 1, 2, 3, 10.
TGOV1N	10	4	Governor records retained only for synchronous machines.
IEEEEX1	10	4	Exciter records retained only for synchronous machines.
IEEEEST	10	4	Stabilizer records retained only for synchronous machines.
PLL1	0	3	PLLs added for the GFL chain at buses 34, 36, 38.
REGCP1	0	3	Converter interface model for GFL resources.
REECA1	0	3	Electrical controller for GFL resources.
REPCA1	0	3	Plant controller for GFL resources.
REGF1	0	3	Grid-forming inverter model at buses 33, 35, 37.
PV	9	9	Static active/reactive dispatch preserved.
Slack	1	1	External/slack generator preserved.

5 Conversion Diagram

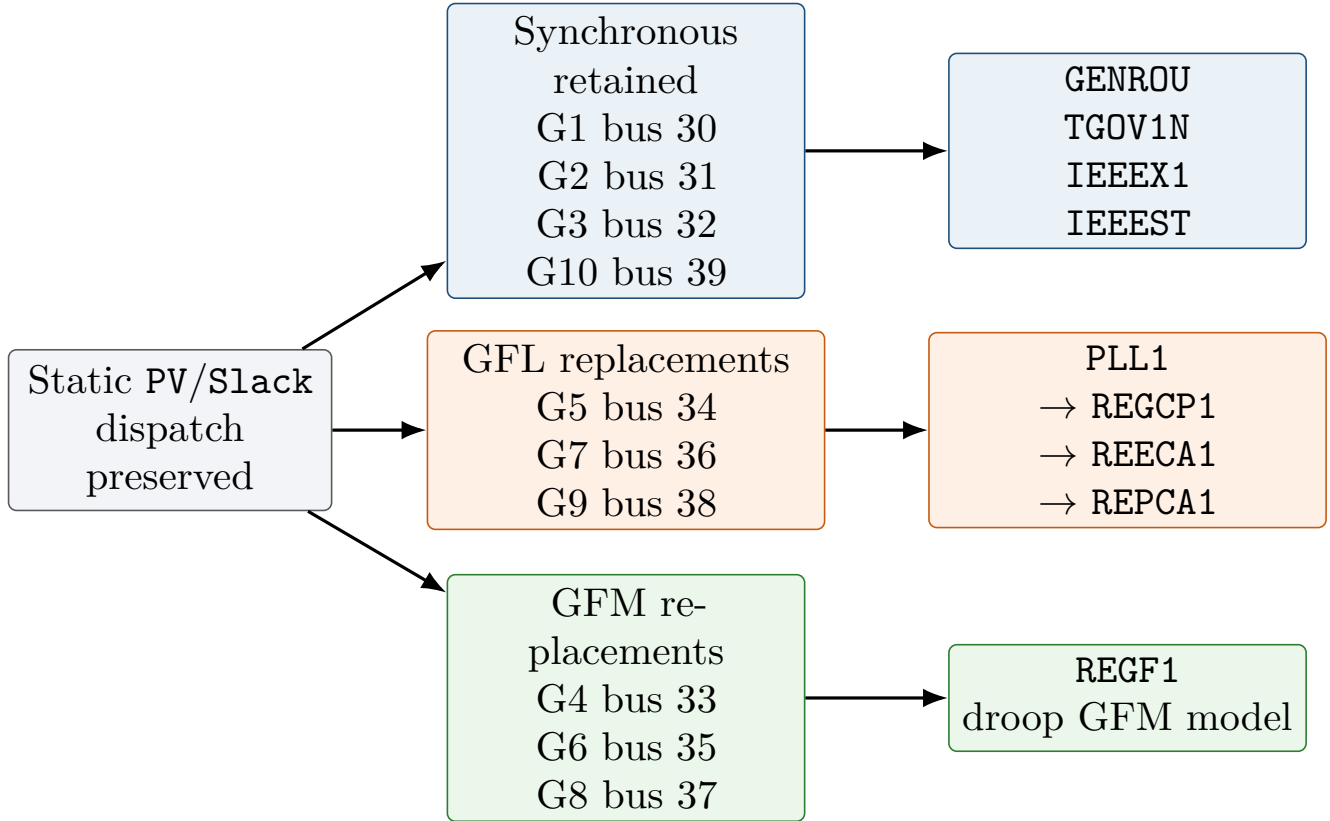


Figure 1: Dynamic model architecture of the mix60 IBR workbook. Static dispatch remains in the PV/Slack sheets; dynamic records are replaced only for the selected generator sites.

6 Variable Provenance

The conversion uses two sources of variables: static generator values from the original PV/Slack rows, and controller parameters declared in `configs/cases/conversion_protocol.yaml`. The table below summarizes the main provenance rules.

New sheet/model	Variables copied from original case	Variables assigned by profile/defaults
PLL1	bus from the converted generator bus.	Kp=1, Ki=0.2, Tf=0.05, Tp=0.05, fn=60.
REGCP1	bus, gen, and Sn; pll links to the new PLL1 record.	Tg=0.1, Rrpwr=10, Brkpt=1, Zerox=0.5, Volim=1.2, Khv=0.7, current limits and gains from <code>published_gfl_baseline</code> .
REECA1	reg links to the new REGCP1; Vref0 and Vref1 use original v0.	Voltage ride-through, current limit, PI gains, and broad limits such as Imax=10, QMax=999, PMAx=999.
REPCA1	ree links to the new REECA1; busf links to the matching BusFreq record.	Plant control flags and gains: VCFlag=1, RefFlag=1, Kp=1, Ki=0.1, Ddn=10, Dup=10.
REGF1	bus, gen, Sn; rf from PV.ra; xf from PV.xs unless overridden; Pmax/Pmin/Qmax/Qmin from static generator limits.	GFM control parameters: wdrp=0.033, Qdrp=0.045, Tr=0.005, Te=0.005, current/voltage PI gains, and Tpm=0.025.

7 Added IBR Records

7.1 Grid-Following Chain

Gen	Bus	PLL	Converter	Electrical control	Plant control
5	34	PLL1_G5	REGCP1_G5	REECA1_G5	REPCA1_G5
7	36	PLL1_G7	REGCP1_G7	REECA1_G7	REPCA1_G7
9	38	PLL1_G9	REGCP1_G9	REECA1_G9	REPCA1_G9

These are configured as GFL because the converter uses REGCP1 with a PLL and a REECA1/REPCA1 control chain. The plant controller uses the corresponding BusFreq record for the generator bus.

7.2 Grid-Forming Records

Gen	Bus	New model	Sn	Pmax	Qmax
4	33	REGF1_G4	1174.8	6.82	5.48208
6	35	REGF1_G6	1085.7	6.97	5.93788
8	37	REGF1_G8	970.2	5.64	4.43468

These are configured as GFM because the dynamic replacement is REGF1, using droop parameters wdrp=0.033 and Qdrp=0.045, with active and reactive limits copied from the original static generator rows.

8 Removed Dynamic Records

For the converted generators 4–9, the synchronous machine and its associated governor, exciter, and stabilizer records are removed:

GENROU removed	TGOV1N removed	IEEEEX1 removed	IEEEEST removed
GENROU_4	TGOV1_4	IEEEEX1_4	IEEEEST_4
GENROU_5	TGOV1_5	IEEEEX1_5	IEEEEST_5
GENROU_6	TGOV1_6	IEEEEX1_6	IEEEEST_6
GENROU_7	TGOV1_7	IEEEEX1_7	IEEEEST_7
GENROU_8	TGOV1_8	IEEEEX1_8	IEEEEST_8
GENROU_9	TGOV1_9	IEEEEX1_9	IEEEEST_9

9 What Did Not Change

- The bus and line network topology is not rewritten by this conversion.
- PV and Slack rows are preserved to keep the same static dispatch and voltage setpoints.
- PQ load rows are not changed by the IBR conversion.
- BusFreq rows are preserved for all generator buses and are linked where needed by REPCA1.
- The disabled original Toggler_1 for GENROU_1 remains disabled in the generated workbook.

10 Important Engineering Notes

1. The conversion is a controlled research transformation, not a canonical official IEEE 39 IBR benchmark.
2. The static power flow is intentionally preserved. Therefore, the IBR version should start from the same solved dispatch as the base case.
3. The GFL parameter set uses broad controller limits, for example $I_{max}=10$ and wide active/reactive limits in REECA1 and REPCA1. These are useful for an initial reproducible build, but they should be tightened or justified before making strong physical claims.
4. The GFM REGF1 records inherit physical limits from the original static generator rows, while their droop and inner-loop gains come from the configured profile.
5. The generated case must still be validated with dynamic gates. A case can be structurally correct and still show questionable voltage or stability behavior under specific contingencies.

11 Reproduction Commands

The IBR workbook can be rebuilt with:

```
$env:SYMPY_GROUND_TYPES='python'
$env:PYTHONPATH='src'
python -m spectral_ibr.interface.cli gate run stage1-build-ibr-case --profile mix60
```

The dynamic comparison can then be run separately with:

```
$env:SYMPY_GROUND_TYPES='python'
$env:PYTHONPATH='src'
python -m spectral_ibr.interface.cli gate run stage1-td-compare --profile mix60
```

12 Summary

The `mix60` workbook is the IBR version of the IEEE 39 dynamic case in which six generator dynamic models are replaced while preserving the static dispatch. Buses 34, 36, and 38 become GFL sites using the `PLL1 -> REGCP1 -> REECA1 -> REPCA1` chain. Buses 33, 35, and 37 become GFM sites using `REGF1`. Buses 30, 31, 32, and 39 remain synchronous. This is the structural change that creates the IBR case; subsequent plots and TDS results are validation of this case, not the definition of the case itself.